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2027 CYCLE

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CUET UG

Mathematics

Financial Mathematics

Perpetuity, Sinking Funds, EMI, Returns, CAGR, Depreciation and Bonds · Section B2 VII · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Perpetuity and sinking funds	"Perpetuity, Sinking Funds: Explain the concept of perpetuity and sinking fund. Calculate perpetuity. Differentiate between sinking fund and saving account." (Section B2 Applied Mathematics Unit VII)	Pricing an endless income; sizing deposits for a future liability.
EMI	"Calculation of EMI: Explain the concept of EMI. Calculate EMI using various methods." (Section B2 Applied Mathematics Unit VII)	Period-rate conversion; reducing-balance EMI computation.
Returns and CAGR	"Calculation of Returns, Nominal Rate of Return: Explain the concept of rate of return and nominal rate of return. Calculate rate of return and nominal rate of return. Compound Annual Growth Rate: Understand the concept of Compound Annual Growth Rate. Differentiate between Compound Annual Growth rate and Annual Growth Rate. Calculate Compound Annual Growth Rate." (Section B2 Applied Mathematics Unit VII)	Return percentages including dividends; CAGR vs AGR distinction.
Linear depreciation	"Linear method of Depreciation: Concept of linear method of Depreciation. Interpret cost, residual value and useful life of an asset from the given information." (Section B2 Applied Mathematics Unit VII)	Equal-slice depreciation schedules; book value year by year.
Bond valuation	"Valuation of Bonds: Concept of bond and related terms. Value of bond using present value approach." (Section B2 Applied Mathematics Unit VII)	Discounting coupons and face value at the market rate.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every

correct answer and −1 for every incorrect answer. The 2026 syllabus closes **Section B2 Applied Mathematics** with this unit — six money formulas, each with a one-line "when does it fire" trigger, and every numeric answer rounding to clean two-decimal rupee figures.

Perpetuity — the Price of an Endless Income

A **perpetuity** is an annuity that never ends: the same payment A, forever, at interest rate i per period. Endless sounds priceless, but the formula is disarmingly small:

$$\text{PV} = \frac{A}{i}$$

Intuition: park a lump sum PV in the bank; the interest it earns each period **is** the payment, while the principal survives untouched. At 8%, a fund of ₹62,500 generates ₹5,000 every year forever. Dividing — never multiplying — is the operation; the "price of forever" is the payment scaled by how hard money works.

WORKED EXAMPLE — PRICING A SCHOLARSHIP FOREVER

A college wants to award a ₹40,000 scholarship every year, indefinitely. The endowment earns 5% per year. How large must the fund be?

Solution. Perpetuity pricing: $\text{PV} = \frac{A}{i} = \frac{40000}{0.05} = 8,00,000$ rupees. The college deposits ₹8,00,000; each year the 5% interest — exactly ₹40,000 — funds the scholarship, and the corpus rides on untouched, ready to pay again next year. If the rate rose to 8%, the same scholarship would cost only $\frac{40000}{0.08} = 5,00,000$ — richer money buys cheaper forever.

Sinking Fund — Saving Up on Purpose

A **sinking fund** flips the annuity: instead of receiving payments, you **deposit** equal amounts so the fund grows to a fixed target exactly when a liability arrives — a debt maturing, machinery due for replacement, a bond issue to retire. With target A, rate i, and n deposits of size P:

$$A = P \cdot \frac{(1+i)^n - 1}{i} \quad \Rightarrow \quad P = A \cdot \frac{i}{(1+i)^n - 1}$$

The syllabus's definitional line — **sinking fund vs savings account** — is a purpose distinction: a savings account holds general money you may spend anytime; a sinking fund is **earmarked**, contractually pointed at one future obligation, with the deposit schedule designed to land exactly on the target.

WORKED EXAMPLE — SIZING THE ANNUAL DEPOSIT

A firm must repay ₹10,00,000 of debentures in 10 years. It sets up a sinking fund earning 8% per year. Find the equal annual deposit.

Solution. $(1.08)^{10} = 2.1589$ (four-decimal table habit). So

$$P = 10,00,000 \cdot \frac{0.08}{(1.08)^{10} - 1} = 80,\frac{000}{1.1589} \approx 69,031 \text{ rupees per year.}$$

Ten deposits of about ₹69,031, each earning 8% along the way, land the fund exactly on ₹10,00,000 — the debenture dies on schedule, no scramble at the end.

EMI — One Payment, Two Jobs

An **equated monthly instalment** is the flat monthly payment that repays a loan's principal **and** interest by the end. Under the reducing-balance method (the standard), interest each month charges only on what remains of the loan. With principal P, **period** rate i, and n instalments:

$$\text{EMI} = P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}$$

Two disciplines make every EMI question workable. First, **convert the rate to the period**: a 12% annual rate is 1% per month — the formula never sees the annual figure raw. Second, know the amortisation story: early EMIs are mostly interest (the outstanding loan is largest), late EMIs are mostly principal — a favourite conceptual MCQ.

WORKED EXAMPLE — EMI END TO END

A loan of ₹1,20,000 at 12% per year, compounded monthly, is repaid in 12 equal monthly instalments. Find the EMI.

Solution. Period rate: $i = 12\frac{\%}{12} = 1\% = 0.01$ per month; $n = 12$. $(1.01)^{12} = 1.1268$. Then

$$\text{EMI} = 1,20,000 \cdot 0.01 \cdot \frac{1.1268}{1.1268 - 1} = 1,\frac{352.16}{0.1268} \approx 10,663 \text{ rupees.}$$

Twelve payments of ₹10,663 total about ₹1,27,956 — roughly ₹7,956 of interest for the year's borrowing. Twelve instalments, one formula, and the monthly-rate conversion doing the quiet heavy lifting.

Rate of Return and the Nominal Rate

The **rate of return** measures how hard money worked, as a percentage of what you put in:

$$\text{Rate of return} = \frac{\text{total gain}}{\text{cost}} \times 100$$

where total gain = selling price − purchase price + every dividend received along the way. Omitting dividends is the classic undercount. The **nominal rate of return** is the **stated** headline rate — before adjusting for inflation or compounding effects; it answers "what did the slip of paper say", not "what did my purchasing power do". Comparing two investments honestly means same period, same basis — the syllabus's point in naming it.

WORKED EXAMPLE — RETURN WITH THE DIVIDEND COUNTED

You buy shares for ₹50,000, receive ₹2,000 in dividends, and sell a year later for ₹55,000. Find the rate of return.

Solution. Total gain: $(55,000 - 50,000) + 2,000 = 7,000$. Rate:

$$\frac{7000}{50000} \times 100 = 14\% \text{ per year.}$$

Capital gain 10%, dividend yield 4% — the sum is the honest 14%. An answer of 10% forgot the dividend; an answer computed on the selling price forgot the base.

Compound Annual Growth Rate — One Rate That Tells the Truth

Growth rarely lands evenly. The **Compound Annual Growth Rate** compresses a start value, an end value, and n years of growth into the single steady rate that would have produced the same ending:

$$\text{CAGR} = \left(\frac{\text{End value}}{\text{Start value}}\right)^{\frac{1}{n}} - 1$$

with n counting **years of growth** between the endpoints. The syllabus's contrast — **CAGR vs Annual Growth Rate (AGR)** — is about volatility: AGR is the plain average of yearly percentage changes; CAGR compounds. They disagree whenever the path zig-zags, and the classic demonstration is: +50% then −50% has AGR 0% but CAGR $\sqrt{1.5 \cdot 0.5} - 1 = \sqrt{0.75} - 1 \approx -29.3\%$ — the money genuinely shrank, and only CAGR says so.

WORKED EXAMPLE — CAGR FROM TWO ENDPOINTS

A business grows from ₹1,00,000 to ₹1,44,000 in 2 years. Find the CAGR.

Solution. $n = 2$ years of growth:

$$\text{CAGR} = \left(\frac{144000}{100000}\right)^{\frac{1}{2}} - 1 = \sqrt{1.44} - 1 = 1.2 - 1 = 0.2 = 20\% \text{ per year.}$$

Whatever the year-by-year wobble, the business earned like a steady 20% compounder. Check: $1,00,000 \cdot 1.2 \cdot 1.2 = 1,44,000$ ✓ — the definition's own verification.

Linear Method of Depreciation — the Straight-Line Slice

Assets wear out. The **linear (straight-line) method** charges the same depreciation every year:

$$D = \frac{\text{Cost} - \text{Residual value}}{\text{Useful life}}$$

— the asset's **cost**, minus whatever it will still be worth at the end (**residual** or scrap value), spread evenly across its **useful life**. The **book value** after k years is Cost $-kD$, a straight line falling to the residual — never below it. The reading skill the syllabus names: identify cost, residual and life from a worded paragraph, then schedule.

WORKED EXAMPLE — A FIVE-YEAR SCHEDULE

A machine costs ₹5,00,000, will be worth ₹50,000 after 5 years. Find the annual depreciation and the book value after 3 years.

Solution. $D = \frac{5{,}00{,}000 - 50{,}000}{5} = 4,50,\frac{000}{5} = 90,000$ per year. After 3 years: book value $= 5,00,000 - 3 \cdot 90,000 = 2,30,000$. The schedule reads like a ladder: 4,10,000 → 3,20,000 → 2,30,000 → 1,40,000 → 50,000 — equal steps of ₹90,000, landing exactly on the residual. Depreciating below ₹50,000 would be a conceptual error, not just arithmetic.

Bond Valuation — the Present-Value Approach

A **bond** is a loan you hold: it promises a **face value** F at **maturity** and fixed **coupons** C (the coupon rate times the face value) until then. Its fair price is the **present value** of everything it will pay, discounted at the **market rate** i — not the coupon rate:

$$V = C \cdot \frac{1 - (1+i)^{-n}}{i} + \frac{F}{(1+i)^n}$$

— an annuity of coupons plus the discounted face value. The moving part is the market rate: coupons are fixed, so when market rates rise above the coupon, the bond's price must fall **below** face (a **discount**) to stay competitive; below the coupon, it trades **above** face (a **premium**).

WORKED EXAMPLE — PRICING A BOND UNDER A RISING MARKET

A bond has face value ₹1,000, an 8% annual coupon, and 3 years to maturity. The market rate is 10%. Value the bond.

Solution. Coupon: $C = 0.08 \cdot 1000 = 80$ per year. Discount factors at 10%: $(1.10)^3 = 1.331$, so $(1.10)^{-3} = 0.7513$.

$$V = 80 \cdot \frac{1 - 0.7513}{0.10} + 1000 \cdot 0.7513 = 80 \cdot 2.487 + 751.3 = 198.96 + 751.3 \approx 950.26 \text{ rupees.}$$

The bond sells at ₹950 — a **discount** to face. Read the logic: why pay ₹1,000 for ₹80 a year when fresh money pays ₹100? The price dips until the bond's yield matches the market. (Market rate 6% would flip it to a premium — try it as practice.)

EVERYDAY EXAMPLE — THE WATER TANK CITY

Think of money as water and rates as tap speeds. A **perpetuity** is a tank so large that opening the tap (interest) pours exactly your daily need forever — the tank size is the daily need divided by the tap speed. A **sinking fund** is filling a tank drop by drop to a marked line before a fixed day. An **EMI** is a drain that empties a loan-tank at one steady rate while interest quietly refills a little of it each month — early on, the refill is fat. **Depreciation** is a leak losing the same depth every year down to the silt line (residual). And a **bond** is buying someone's incoming drip system at today's water prices: if the city now sells water cheaper (market rates up), their drip system is worth less than its label.

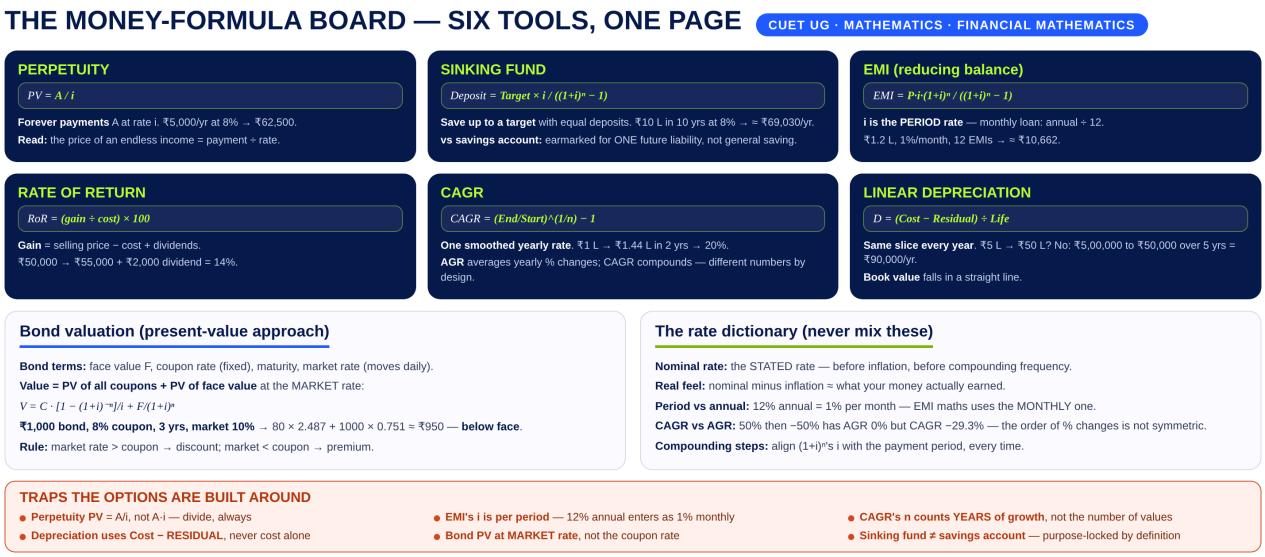


Figure 1: The six money tools on one board: perpetuity, sinking fund, EMI, return, CAGR, depreciation — plus the bond's present-value engine and the rate dictionary.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — FINANCIAL MATHEMATICS

Formula / rule	Letters mean	Fire it when
Perpetuity: $\text{PV} = \frac{A}{i}$	endless payment A, rate i	forever-income pricing
Sinking fund deposit: $P = A \cdot \frac{i}{(1+i)^n - 1}$	target A in n deposits	future liability
Sinking fund vs savings account: purpose-locked vs general	definition	contrast MCQs
Period rate: annual ÷ 12 for monthly	EMI's i	all EMI sums
$\text{EMI} = P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}$	reducing balance	loan instalments
Early EMIs: mostly interest	amortisation fact	concept MCQs
Rate of return $= \frac{\text{total gain}}{\text{cost}} \times 100$	gain includes dividends	return questions
Nominal rate: stated, before inflation	headline rate	definition MCQs

Formula / rule	Letters mean	Fire it when
CAGR $= \left(\frac{\text{End}}{\text{Start}}\right)^{\frac{1}{n}} - 1$	one smoothed yearly rate	two-endpoint growth
AGR = plain average of yearly % changes	volatile paths disagree with CAGR	contrast stems
Linear depreciation: $D = \frac{\text{Cost} - \text{Residual}}{\text{Life}}$	equal slices	schedules
Book value after k years: Cost $-kD$	never below residual	year-by-year
Bond: $V = C \cdot \frac{1-(1+i)^{-n}}{i} + \frac{F}{(1+i)^n}$	coupons + face, at MARKET rate	valuation
Market rate $>$ coupon $\rightarrow$ discount; $<$ $\rightarrow$ premium	fixed coupons, moving market	price direction

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — FINANCIAL MATHEMATICS

The mistake	What is right	The trap it sets
Perpetuity priced as $A \cdot i$	PV $= \frac{A}{i}$ — divide	Multiplied-perpetuity options
Sinking fund treated as ordinary saving	Earmarked for one liability	Definition-swap options
Annual rate used raw in EMI	Convert to the monthly period rate first	Inflated EMI numerics
Dividends omitted from returns	Total gain includes every payout	Understated return percentages
CAGR with n = number of values	n counts years of GROWTH	Off-by-one exponents
AGR quoted as CAGR	Volatile paths separate the two	0%-for-−29.3% lures
Depreciation on full cost	Cost minus RESIDUAL, then slice	Oversized depreciation rows
Bond discounted at the coupon rate	The MARKET rate drives present value	Face-value-priced options

Sixty-Second Self-Test

Statement	Verdict
A perpetuity's price falls when interest rates rise.	True — $\frac{A}{i}$ shrinks as i grows
A sinking fund is just a savings account with a nicer name.	False — purpose-locked to a liability
A 12% annual loan enters EMI maths as 1% monthly.	True — period conversion first
CAGR of +50% then −50% is 0%.	False — that is AGR; CAGR is about −29.3%
Depreciation uses cost minus residual value.	True — the residual is protected
A bond trades above face when market rates exceed its coupon.	False — that is a discount situation

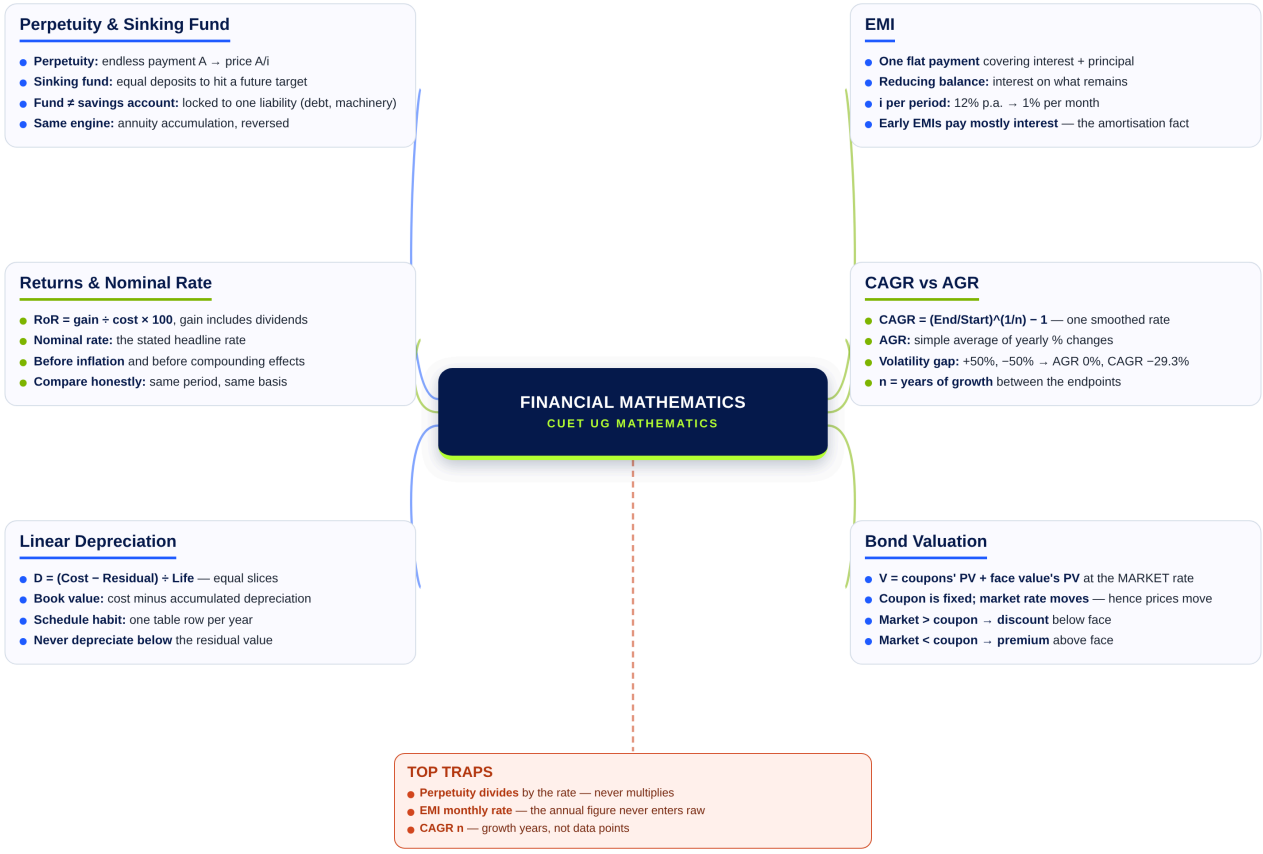


Figure 2: The chapter in one look: six money tools, the rate dictionary, and the traps each one plants.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Name the tool first — perpetuity, fund, EMI, return, CAGR, depreciation or bond — then compute as little as possible.

Q1. A trust wants to give ₹6,000 every year forever from a fund earning 6% per year. The required fund size is:

(A) ₹1,00,000 **(B)** ₹36,000
(C) ₹1,06,000 **(D)** ₹60,000

Q2. A sinking fund differs from a savings account because it:

(A) earns no interest **(B)** is earmarked for one specific future liability
(C) cannot be withdrawn **(D)** pays tax monthly

Q3. A loan charges 9% per year, compounded monthly. The monthly rate used in EMI calculations is:

(A) 9% **(B)** 0.75%
(C) 1% **(D)** 0.09%

Q4. For principal P, period rate i and n instalments, the reducing-balance EMI equals:

(A) $P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n}$ **(B)** $P \cdot \frac{(1+i)^n}{i}$
(C) $P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}$ **(D)** $\frac{P}{n} + P \cdot i$

Q5. Shares bought for ₹20,000 are sold for ₹21,600 with no dividends. The rate of return is:

(A) 6% **(B)** 8%
(C) 10% **(D)** 16%

Q6. A business grows from ₹2,00,000 to ₹2,88,000 in 2 years. Its CAGR is:

(A) 44% **(B)** 12%
(C) 20% **(D)** 24%

Q7. An investment rises 50% one year and falls 50% the next. Its CAGR is about:

(A) −29.3% **(B)** 0%
(C) 25% **(D)** −50%

Q8. A machine costs ₹8,00,000 with residual value ₹80,000 and a 4-year life. Annual linear depreciation is:

(A) ₹2,00,000 **(B)** ₹80,000
(C) ₹1,60,000 **(D)** ₹1,80,000

Q9. For the same machine, its book value after 2 years is:

(A) ₹5,60,000 **(B)** ₹6,20,000
(C) ₹4,40,000 **(D)** ₹4,00,000

Q10. A bond's coupon rate is 7% while the market rate is 9%. The bond trades:

(A) above face value **(B)** at face value
(C) below face value **(D)** at the coupon price

Detailed Solutions

Sol. Q1. Perpetuity pricing: PV $= \frac{A}{i} = \frac{6000}{0.06} = 1,00,000$ rupees. At 6%, one lakh of principal throws off exactly ₹6,000 forever. Option (B) multiplied instead of dividing.

Answer: (A)

Sol. Q2. By definition, a sinking fund is earmarked for one named future liability — a debenture repayment, a machine replacement — with deposits designed to land on the target. Interest, withdrawals and tax are side issues, not the distinction.

Answer: (D)

Sol. Q3. Period conversion: $9\% \div 12 = 0.75\%$ per month. Option (C) reflex-divided by 9; option (A) used the annual rate raw — the two standard EMI-ruining slips.

Answer: (B)

Sol. Q4. The reducing-balance EMI formula: $P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}$. Option (A) lost the "−1" — without it, no loan ever finishes paying.

Answer: (C)

Sol. Q5. Gain: $21600 - 20000 = 1600$. Rate: $\frac{1600}{20000} \times 100 = 8\%$. No dividend to add — the clean capital-gain-only case.

Answer: (B)

Sol. Q6. CAGR $= \left(\frac{288000}{200000}\right)^{\frac{1}{2}} - 1 = \sqrt{1.44} - 1 = 0.2 = 20\%$ per year. Check: $2,00,000 \cdot 1.2 \cdot 1.2 = 2,88,000$ ✓. Option (A) read the total growth as the yearly rate.

Answer: (C)

Sol. Q7. End/start $= 1.5 \cdot 0.5 = 0.75$, so CAGR $= \sqrt{0.75} - 1 \approx -0.293 = -29.3\%$. The 0% option is the AGR answer — the average of +50 and −50 — and it hides a real loss.

Answer: (A)

Sol. Q8. Depreciable base: $800000 - 80000 = 7,20,000$. Annual slice: $\frac{720000}{4} = 1,80,000$ rupees. Option (A) forgot the residual — the most repeated depreciation slip.

Answer: (D)

Sol. Q9. Book value after 2 years: $800000 - 2 \cdot 180000 = 800000 - 360000 = 4,40,000$ rupees. The ladder falls by exactly one slice per year, never below the residual.

Answer: (C)

Sol. Q10. Market rate 9% beats the coupon 7%: fresh money pays more, so this bond's price dips below face until its yield matches — a discount. Above-face pricing happens only when the coupon wins.

Answer: (B)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(B) in Q1	Computed the interest on a guessed fund	Perpetuity divides: $\frac{A}{i}$ — say the operation aloud
(A) in Q3	Divided by 9 instead of 12	Months, not instalment myths — annual ÷ 12
(A) in Q6	Quoted total growth as yearly	Two years of growth means the square root, not the total
(B) in Q7	Reached for AGR reflexes	Averages of % changes lie on volatile paths — compound it
(A) in Q8	Depreciated the full cost	Residual value is protected — subtract it first
(A) in Q10	"Fixed coupon = fixed price"	Coupons are fixed; the MARKET rate prices the bond

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